Mahesh Sreekumar Rajasree[1][2]

CISPA Helmholtz Center for Information Security
Saarbrücken
Saarland, Germany

PRINCIPAL INTERESTS Cryptography, Quantum Cryptography, Lattice theory, Algorithms, Cryptanalysis

Employment

Post Doctoral Researcher December 2024 - Present
Host: Prof. Nico Döttling
CISPA Helmholtz Center for Information Security
Saarbrücken, Saarland, Germany

Post Doctoral Fellow April 2023 - November 2024
Supervisor: Prof. Venkata Koppula
Computer Science and Engineering
Indian Institute of Technology Delhi, New Delhi, India

ACADEMIC BACKGROUND

Ph.D. & M.Tech. Computer Science and Engineering August 2016 - July 2023
Indian Institute of Technology Kanpur, Uttar Pradesh, India

- Advisor: Prof. Manindra Agrawal
- Ph.D. Thesis: Analysis of Symmetric-Key Cryptosystems and Subset-Sum Problem
- M.Tech. Thesis: New Results and Proofs in Lattice Theory
- CPI: 10/10

B.Tech. Computer Science and Engineering August 2012 - May 2016
National Institute of Technology Calicut, Kerala, India

- Advisor: Dr. Sumesh T.A.
- B.Tech. Thesis: Firewall for Preventing Anonymous Proxy Usage
- CPI: 8.89/10

Secondary & Higher Secondary Education June 2009 - March 2012
Arya Central School, Trivandrum, Kerala, India

- All India Senior Secondary School Examination Percentage: 91.2
- All India Secondary School Examination CGPA: 9.6/10

Peer-reviewed Works

1. **Separating Pseudorandom Codes from Local Oracles**
   with Nico Döttling and Anne Müller
   23rd Theory of Cryptography, TCC 2025
   Outstanding Paper Award
2. **Incompressible Encryption Beyond CPA/CCA Security**
   with Venkata Koppula, Abhinav Kumar and Harihar Swaminathan
   Communications in Cryptology, Volume 2, Issue 2, 2025

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3. **A Note on Adaptive Security in Hierarchical Identity-Based Encryption**
   with Rishab Goyal and Venkata Koppula
   45th Annual International Cryptology Conference, CRYPTO 2025
4. **Non-Committing Identity Based Encryption: Constructions and Applications**
   with Rishab Goyal, Fuyuki Kitagawa, Venkata Koppula, Ryo Nishimaki and Takashi Yamakawa
   28th IACR International Conference on Practice and Theory of Public-Key Cryptography, PKC 2025
5. **Incompressible Functional Encryption**
   with Rishab Goyal, Venkata Koppula and Aman Verma
   16th Innovations in Theoretical Computer Science, ITCS 2025
6. **Leakage-Resilient Key-Dependent Message Secure Encryption**
   with Dhairya Gupta and Harihar Swaminathan
   25th International Conference on Cryptology, INDOCRYPT 2024
7. **Leakage-Resilient Incompressible Cryptography: Constructions and Barriers**
   with Kaartik Bhushan, Rishab Goyal, Venkata Koppula, Varun Narayanan and Manoj Prabhakaran,
   International Conference on the Theory and Application of Cryptology and Information Security, ASIACRYPT 2024
8. **Complexity of Monomial Prediction in Cryptography and Machine Learning**
   with Pranjal Dutta and Santanu Sarkar,
   26th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing, SYNACS 2024
9. **A New Reduction among Lattice Problems**
   with Rajendra Kumar and Shashank K Mehta,
   26th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing, SYNACS 2024
10. **Efficient reductions and algorithms for Subset Product**
    with Pranjal Dutta,
    9th Annual International Conference on Algorithms and Discrete Applied Mathematics, CALDAM 2023
    Best Student Presentation Award
11. **On the Variants of Subset Sum: Projected and Unbounded**
    with Pranjal Dutta,
    25th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing, SYNACS 2023
12. **On the bases of $\mathbb{Z}^n$ lattice**
    with Shashank K Mehta,
    24th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing, SYNACS 2022
13. **Weak-keys and key-recovery attack for TinyJAMBU**
    with Pranjal Dutta and Santanu Sarkar,
    Scientific Report, Nature
14. **On the hardness of monomial detection and zero-sum distinguishers for Ascon**

with Pranjal Dutta and Santanu Sarkar,
12th International Workshop on Coding and Cryptography, WCC 2022

15. **Algebraic algorithms for variants of Subset Sum problem**
with Pranjal Dutta,
8th Annual International Conference on Algorithms and Discrete Applied Mathematics, CALDAM 2022
Best Student Presentation Award

16. **Cryptanalysis of Round-Reduced KECCAK using Non-Linear Structures**
20th International Conference on Cryptology, INDOCRYPT 2019

17. **Cryptanalysis of 1-Round KECCAK**
with Rajendra Kumar and Hoda Al Khzaimi,
10th International Conference on Cryptology, AFRICACRYPT 2018

Unpublished works (Under submission)

1. **Weak Pigeonhole Equal-Sums made Simpler and Faster**
with Deepak Bhati, Pranjal Dutta, Antoine Joux and Karol Węgrzycki
2. **Quantum Pseudorandom Codes**
with Nico Döttling and Anne Müller
3. **Chosen Ciphertext Secure Pseudorandom Codes from Standard Assumptions**
with Nico Döttling, Venkata Koppula and Hendrik Waldner

Academic Achievements

1. Won the Outstanding Paper award in TCC 2025.
2. Won the Best Student Presentation award (one of the 5 winners) in CALDAM 2023.
3. Won the Best Student Presentation award (one of the 3 winners) in CALDAM 2022.
4. Awarded **Prime Minister's Research Fellowship** in 2018 (received approximately USD 50,000 for a duration of 4 years to cover stipend and other expenses related to research).
5. Awarded Academic Excellence 2017 at IIT Kanpur.
6. Secured All Indian Rank 28 in Joint Entrance Screening Test (JEST-2016).
7. Secured All India Rank 575 in Graduate Aptitude Test in Engineering (GATE 2016)
8. Secured fourth place in the first round of Amrita InCTF 2015.

Short Term Research Visit

1. **Technical University of Denmark** (Dec 2025), Host: Prof. Christian Majenz
2. **International Institute of Information Technology Hyderabad** (2024), Host: Dr. Suryajith Chillara
3. **International Centre for Theoretical Sciences Bangalore** (Oct 2023), Host: Prof. Jaikumar Radhakrishnan
4. **Indian Institute of Technology Bombay** (Jul 2023), Host: Prof. Manoj M. Prabhakaran
5. **Institute of Science and Technology Austria** (Sept 2022), Host: Prof. Krzysztof Pietrzak

Research Internships & Projects

1. Visiting scholar at IIT Delhi hosted by Prof. Venkata Koppula (Feb 2023).
2. Visiting scholar at Microsoft Research India hosted by Satya Lokam (Jun 2018, Oct 2018, Mar 2019, Sept 2019).
3. Visiting scholar at New York University, Abu Dhabi hosted by Prof. Hoda AlKhzaimi (Oct 2017).

Academic Talks and Presentations

1. **CRYPTO 2025**, Conference Talk.
   Title: A Note on Adaptive Security in Hierarchical Identity-Based Encryption
2. **PKC 2025**, Conference Talk.
   Title: Non-Committing Identity Based Encryption: Constructions and Applications
3. **ITCS 2025**, Conference Talk.
   Title: Incompressible Functional Encryption
4. **INDOCRYPT 2024**, Conference Talk.
   Title: Leakage-Resilient Key-Dependent Message Secure Encryption
5. **SYNACS 2024**, Conference Talk.
   Title: Complexity of Monomial Prediction in Cryptography and Machine Learning
6. **SYNACS 2024**, Conference Talk.
   Title: A New Reduction among Lattice Problems
7. **Invited Talk**, CSTAR, IIIT Hyderabad.
   Title: Leakage-Resilience meets Incompressibility
8. **Invited Talk**, University of Wollongong, Australia.
   Title: Incompressible Encryption beyond Public Key Encryption
9. **Invited Talk**, IIST, Trivandrum.
   Title: Incompressible Encryption beyond Public Key Encryption
10. **Invited Talk**, Microsoft Research, Bangalore.
    Title: Incompressible Encryption beyond Public Key Encryption
11. **SYNACS 2023**, Conference Talk.
    Title: On the Variants of Subset Sum: Projected and Unbounded
12. **Invited Talk**, Trust Lab, IIT Bombay.
    Title: Incompressible Encryption
13. **CALDAM 2023**, Conference Talk.
    Title: Efficient reductions and algorithms for Subset Product
14. **SYNACS 2022**, Conference Talk.
    Title: On the bases of $\mathbb{Z}^n$ lattice
15. **SIGTACS**, CSE, IIT Kanpur.
    Title: On the hardness of monomial prediction problem.
16. **Invited Talk**, CSE, IUST Jammu & Kashmir.
    Title: On the Subset Sum Problem.
17. **SIGTACS**, CSE, IIT Kanpur.
    Title: An Algebraic Algorithm for Hamming Subset-Sum Problem.
18. **WCC 2022**, Workshop Talk.
    Title: On the hardness of monomial prediction problem.
19. **Invited Talk**, PMRF Symposium.
    Title: Cryptanalysis of KECCAK & Algorithms for Lattice Problems.

20. **SIGTACS**, CSE, IIT Kanpur.
    Title: Cryptanalysis of Round-Reduced KECCAK using Non-linear Structures.
21. **INDOCRYPT 2019**, Conference Talk.
    Title: Cryptanalysis of Round-Reduced KECCAK using Non-Linear Structures.
22. **SIGTACS**, CSE, IIT Kanpur.
    Title: Leftover Hash Lemma
23. **SIGTACS**, CSE, IIT Kanpur.
    Title: Lattices in Computer Science
24. **SIGTACS**, CSE, IIT Kanpur.
    Title: A Deterministic Exponential Time Algorithm for some lattice problems

Teaching Experience

1. CRYPTO25 - Cryptography at Saarland University, 2025. Delivered a week-long lecture series on key agreement protocols and public-key encryption.
2. COL100 - Introduction to Computer Science at IITD, 2024. Delivered lectures, designed examinations, and supervised laboratory sessions, covering approximately one-third of the course.
3. COL863 - Special Topics in Theoretical Computer Science at CSE, IITD, 2024. Delivered a module on lattices, designed assignments, and graded coursework.

Supervision

1. PhD mentored:
   (a) Deepak Bhati, CISPA
   (b) Hadi Barzegar, CISPA
   (c) Anne Müller, CISPA
2. Master theses mentored:
   (a) Abhinav Kumar, IIT Delhi (co-authored paper in Communications in Cryptology, 2025)
   (b) Dhairya Gupta, IIT Delhi (co-authored paper at INDOCRYPT 2024)
3. Bachelor projects supervised
   (a) Naman Verma, IIT Kanpur.
4. Bachelor projects mentored:
   (a) Anish Banerjee, IIT Delhi
   (b) Shankh Gupta, IIT Delhi
   (c) Harihar Swaminathan, IIT Delhi (co-authored paper in Communications in Cryptology, 2025 and INDOCRYPT 2024)

Teaching Assistance

1. COL872 - Special Topics in Cryptography at CSE, IITD, 2023
2. CS961 - Modern Cryptology at Emaster Program, IITK 2023, 2022
3. CS972 - Introduction to Linear Algebra at Emaster Program, IITK 2022
4. Modern Algebra at NPTEL 2021, 2018
5. CS641 - Modern Cryptology at CSE, IITK 2021, 2020, 2019, 2018, 2017
6. CS345 - Algorithms II at CSE, IITK 2018
7. ESO207 - Data Structures and Algorithms at CSE, IITK 2019, 2017
8. CS3092D - Operating Systems Laboratory at CSE, NITC 2016

Professional Activities

1. Reviewed papers for INDOCRYPT 2021, MFCS 2023, FSTTCS 2023, Journal of Cryptology, EUROCRYPT 2023, ASIACRYPT 2024, ITCS 2025, EUROCRYPT 2025, PKC 2025, ICALP 2025, CRYPTO 2025, STOC 2026, ICALP 2026.
2. Part of the Indian team for the Australian-India Cyber and Critical Technology Partnership (AICCTP). The partnership involves activities that facilitate Australia-India research partnerships to strengthen understanding of ethical frameworks for emerging and critical technologies, and the development of best practices on translating ethical frameworks into practical action and initiatives that encourage the development of technical critical technology standards that advance Australia and India's interests.

Volunteer Works

1. Lecture delivered at Continuing Education Programme for employees at Centre For Railway Information Systems, IIT Delhi 2024.
2. Lecture delivered at STEM Mentorship Program for High School Girl Students, IIT Delhi 2023.
3. Volunteer at Foundations of Software Technology and Theoretical Computer Science, FSTTCS 2017.
4. Volunteer at Cybersecurity Awareness Week CSAW 2017.
5. Head Coordinator of Special Interest Group on Theoretical Aspects of Computer Science SIGTACS at IIT Kanpur.

Conference & Workshop Attended (offline)

1. 23rd International Conference on Theory of Cryptography, TCC 2025, Aarhus, Denmark.
2. 45th Annual International Cryptology Conference, CRYPTO 2025, Santa Barbara, CA, USA.
3. Cryptography 10 Years Later: Obfuscation, Proof Systems, and Secure Computation at Simons Institute for the Theory of Computing (2025), Berkeley, USA.
4. 28th IACR International Conference on Practice and Theory of Public-Key Cryptography, PKC 2025, Røros, Norway.
5. 16th Innovations in Theoretical Computer Science Conference, ITCS 2025, Columbia University, New York, USA.
6. 22nd International Conference on Theory of Cryptography, TCC 2024, Milan, Italy.
7. Warsaw IACR Summer School on Post-Quantum Cryptography 2024, Warsaw, Poland.
8. 9th Annual International Conference on Algorithms and Discrete Applied Mathematics, CALDAM 2023, Gandhinagar, India.
9. 24th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing, SYNACS 2022, Hagenberg, Austria.
10. Quantum Algorithms at Simons Institute for the Theory of Computing (2020), Berkeley, USA.
11. Quantum Cryptanalysis of Post-Quantum Cryptography at Simons Institute for the Theory of Computing (2020), Berkeley, USA.
12. Lattices: Geometry, Algorithms and Hardness at Simons Institute for the Theory of Computing (2020), Berkeley, USA.

13. Secure Multiparty Computation: Theory and Practice workshop held at IISc (2020), Bangalore, India.
14. 20th International Conference on Cryptology, INDOCRYPT 2019, Hyderabad, India.
15. Theory of Cryptography Conference, TCC 2018, Goa, India.
16. 37th Foundations of Software Technology and Theoretical Computer Science 2017, Kanpur, India.

Conference & Workshop Attended (online)

1. 26th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing, SYNACS 2024, Timișoara, Romania.
2. 25th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing, SYNACS 2023, Nancy, France.
3. 12th International Workshop on Coding and Cryptography, WCC 2022.
4. 8th Annual International Conference on Algorithms and Discrete Applied Mathematics, CALDAM 2022

Referees

1. **Prof. Nico Döttling**
   CISPA Helmholtz
   Email: doettling@cispa.de
2. **Prof. Antoine Joux**
   CISPA Helmholtz
   Email: joux@cispa.de
3. **Prof. Venkata Koppula**
   Department of CSE, IIT Delhi
   Email: kvenkata@cse.iitd.ac.in
4. **Prof. Manoj Prabhakaran**
   Department of CSE, IIT Bombay
   Email: mp@cse.iitb.ac.in
5. **Prof. Manindra Agrawal**
   Department of CSE, IIT Kanpur
   Email: manindra@cse.iitk.ac.in
6. **Prof. Shashank K Mehta**
   Department of CSE, IIT Kanpur (Retired)
   Email: skmehta@cse.iitk.ac.in